

NGUYEN DOAN HIEU NGUYEN

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SUMMARY

Nguyen is a Combined Master's & Ph.D. student in Integrative Biotechnology at Sungkyunkwan University, with a strong academic foundation as the Valedictorian in Artificial Intelligence from FPT University, Vietnam. His interests center on deep learning approaches for biological data. He excels in research design, coding, and proposing novel ideas, with a primary focus on computational methods. His long-term goal is to develop innovative deep learning frameworks that contribute meaningfully to computational biology.

Research interest: Computational biology, Bioinformatics, Deep Learning.

EDUCATION

Sungkyunkwan University (SKKU)

Combined Master's & Ph.D. of Integrative Biotechnology

Sep 2024 – Now

Suwon-si, Republic of Korea

FPT University - Graduated as Valedictorian

Bachelor of Information Technology (Artificial Intelligence) - GPA 9.2/10

Sep 2020 – March 2024

Ho Chi Minh City, Vietnam

PUBLICATIONS

- **Nguyen, N. D. H.**, Pham, N. T., Seo, H., Wei, L., & Manavalan, B. (2025). Mulaqua: An Interpretable Multimodal Deep Learning Framework for Identifying PMT/vPvM substances in Drinking Water. *Journal of Hazardous Materials*, 140573. <https://doi.org/10.1016/j.jhazmat.2025.140573> 🌐
- **Nguyen, N. D. H.**, Pham, N. T., Tran, D. T., Wei, L., Malik, A., & Manavalan, B. (2025). xBitterT5: an explainable transformer-based framework with multimodal inputs for identifying bitter-taste peptides. *Journal of Cheminformatics*, 17(1), 127. <https://doi.org/10.1186/s13321-025-01078-1> 🌐
- Tran, D. T., **Nguyen, N. D. H.**, Pham, T. T., Tran, P., Vu, T. T., Nguyen, C. T., Dang-Ngoc, H., & Dang, D. N. M. (2024). SwinTEExCo: Exemplar-based video colorization using Swin Transformer. *Expert Systems With Applications*, 260, 125437. <https://doi.org/10.1016/j.eswa.2024.125437> 🌐
- **Nguyen, N.**, Pham, N. T., Tran, D., & Manavalan, B. (2024). Lang2Mol-Diff: A Diffusion-Based Generative Model for Language-to-Molecule Translation Leveraging SELFIES Representation. *Proceedings of the 1st Workshop on Language + Molecules (L+M 2024)* (Pp. 128–134). Bangkok, Thailand: Association for Computational Linguistics, 128–134. <https://doi.org/10.18653/v1/2024.langmol-1.15> 🌐

AWARDS

- Awarded Champion prize at FPT Education Research Festival 2023 - Competition on Scientific Research 🌐
- Awarded Champion prize at FPT Education Hackathon 2022 - Competition on Blockchain application 🌐
- Awarded FPT University Golden Toad of Engineering Major Spring 2022 - the award for the most excellent student in a semester 🌐
- Awarded Third prize of Algebra group in FPT University Mathematics Olympic 2022 🌐
- Achieved 100% Scholarship at FPT University HCM 🌐

EXPERIENCE

Computational Biology and Bioinformatics Laboratory - SKKU 🌐

Graduate Researcher

Sep 2024 – Now

Suwon-si, Republic of Korea

AI Technology and Application Research Lab - FPTU 🌐

Undergraduate Researcher

Jun 2023 – Dec 2024

Ho Chi Minh City, Vietnam

SKILLS

Research skills: analyzing, data collection, critical thinking

Technical skills: Python, deep learning, Ubuntu

CERTIFICATIONS

- Deep Learning
- Generative Adversarial Networks (GANs)
- TensorFlow: Advanced Techniques

LANGUAGES

- **Languages:** English (Professional working proficiency), Vietnamese (Native language)